Jaya Sharma

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Introduction

I'm a PhD student in Cybersecurity Engineering at the Technical University of Denmark (DTU), working under the supervision of Dr. Christian Majenz. My PhD research lies at the intersection of theoretical cryptography and quantum computing, with a focus on the security of cryptographic constructions in the quantum random oracle model- so far I have worked on security non-interactive zero knowledge proofs and post-quantum threshold signatures based on hardness of group based assumptions. My other specific interests include lattice-based cryptography, quantum information and error correction.

Employment

Technical University of Denmark (DTU) 2024-27
PhD Student at Cybersecurity Engineering Section, DTU Compute
Supervisor: Dr. Christian Majenz

Indraprastha Institute of Information Technology Delhi (IIITD) 2022-24
PhD Junior Research Fellow in Department of Computer Science and Engineering
Area:Lattice Based Cryptography and Quantum Computing.

Education

Indian Institute of Technology Roorkee (IITR) 2020-22
Master of Science in Mathematics Grade: 8.675/10

University of Delhi- Lady Shri Ram College For Women (LSR) 2017-20
Bachelor of Science (Honors) in Mathematics Grade: 9.081/10

Research Visits/Internships

Centrum Wiskunde & Informatica (CWI) 09/2026-01/2027
Host: Dr. Serge Fehr
-Working on security proofs for post-quantum non interactive zero knowledge protocols.

QWorld Association-QIntern *(Remote internship)* 07-09/2025
Project: Implementing Efficient block-encoding of dense matrices via QRAM.

Indian Institute of Science, Bangalore (IISc) 06-07/2023
Mentor: Dr. Chaya Ganesh
-Studied compression theory of sigma protocols and lattice based sigma protocols.

Indian Institute of Science Education and Research (IISER) Bhopal*(Remote internship)* 06-08/2021
Mentor: Dr. Sanjay Singh
Project:Introduction to Algebraic Function Fields and Codes

Publications

Security of the Fischlin's Transform in Quantum Random Oracle Model 2026
QCrypt'26, IACR ASIACRYPT'26
Christian Majenz, Jaya Sharma

Projects

Implementating Efficient block-encoding of dense matrices via QRAM using the Qrisp framework 2025
QWorld QIntern Project 2025(Library currently in review by Qrisp Developers)
Viraj Dsouza, Kalpesh Bhatnagar, Kyle Carlo Lasala, Jaya Sharma

A Survey on Quantum Error Correcting Codes 2023

Elliptic Curves Over Complex Numbers 2022
Master's Thesis
Supervisor- Dr. A. Swaminathan

Brent-Dekker Method: An Attempt to Decode 2020
Eclat- A Mathematics Journal, Vol 11
Dr. Monika Singh, Shagun Agrawal, Jaya Sharma

Presentations

Security of the Fischlin Transform in Quantum Random Oracle Model- QCrypt UOttawa 08/2026
Fiat Shamir Transform in QROM- Crypto Reading Group DTU 10/2024
Lattice Based Sigma Protocols- Theory Reading Group IIITD 10/2023
Trapdoors for Hard Lattices and New Crytographic Constructions-Theory Reading Group IIITD 05/2023
The Rabin Cryptosystem: Mathematical Foundations and Implementation- IIITD 12/2022
Coherent Measures of Financial Risk- MSc Mathematics Seminar, IITR 10/2021
Elliptic Curve Cryptography- Department of Mathematics, LSR 05/2019
Latin Squares- Recreation, Disproving Euler to Cryptography- Department of Mathematics, LSR 05/2018
Applications of Congruence Modulo- Department of Mathematics LSR 02/2018

Honours, Awards & Grants

Asiacrypt Outstanding Paper Award 2026

Thomas B. Thriges Fond Grant, William Demant Fonden Grant, Otto Mønsteds Fond Grant 2026
-secured three Danish Grants to fund research stay at CWI Amsterdam.

ACM-India Anveshan Setu Fellowship 2023
- Awarded by Association for Computing Machinery(ACM) India to work at IISc.

Joint CSIR-UGC Junior Research Fellowship-All India Rank 122/30942 candidates 2022
-Awarded PhD Research Fellowship in Mathematical in the exam conducted jointly by Council for Scientific and Industrial Research and University Grants Commission India.

TIFR Vigyan Vidushi Fellow- Mathematics and Theotetical Computer Science 2021
- Selected among few female STEM students from all over India for two discipline-specific summer schools organised by the Tata Institute of Fundamental Research (TIFR), Mumbai.

IIT JAM Mathematics: All India Rank 143/14374 candidates 2020
- Joint Admission Test for Master of Science Courses at Centrally Funded Technical Institutions in India.

Other Activities/Service

Sub-Reviwer Eurocrypt'27 2026
PhD Committee Member- DTU Compute 2025

Teaching Assistance
-Applied Cryptography , Discrete Mathematics , Graph Theory (DTU) 2024-25
-Introduction To Cryptography (IIT Delhi) 2023
-Discrete Mathematics, (IIITD) 2022

Illustrator- LSR College Magazine 2019
Event Coordinator- Enigma, Annual Academic Meet Department of Mathematics LSR 2019

Programming

C++, Python, Qiskit.

Workshops/Summer Schools/Conferences

Qcrypt- Ottawa, Canada 08/2026
Eurocrypt- Rome, Italy 05/2026
DQC Scientific Quantum Conference 2026- Aalborg University, Denmark 04/2026
Theory of Cryptography Conference (TCC) 2025-Aarhus, Denmark 12/2025
Workshop on Mathematics of Post-Quantum Cryptography- University of Zurich 06/2025
Quantum Safe Internet (QSI) Workshop- Technical University of Denmark, Lyngby 05/2025
IACR Summer School on Post Quantum Cryptography- University of Warsaw 07/2024
ACM India Grad Cohort- IISc Bangalore 07/2023
CSE Research Symposium- IIT Bombay 03/2023

Vigyan Vidushi Mathematics Summer School- TIFR Mumbai	07/2021
STCS Vigyan Vidushi Summer School on Theoretical Computer Science- TIFR Mumbai	06/2021
Virtual Math Fest India	07/2020
Workshop: Calculus the Geometrical Way- Miranda House College, University of Delhi	09/2019